# CodeAug: Architectural Master Plan & Engineering Specification

**Project:** Remote Sensing Image Change Captioning (RSICC) — Indian Urban & Seasonal Domain Adaptation
**Target Infrastructure:** ADA Cluster (NVIDIA GTX 1080 Ti 11 GB VRAM & CPU-Only Compute Nodes)
**Codebase Alignment:** Fully compliant with src/ modular pipeline (Phase 5/6 & src.dataset.py)

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## 1. Executive Summary & Base Project Context

This formal engineering document establishes the upgraded methodology for our Remote Sensing Image Change Captioning (RSICC) pipeline. Our project base in src/ evolved from a baseline two-stream CNN (RSICCformerBaseline) to Phase 5 (RemoteCLIPCrossAttentionModel) and Phase 6 (TileBasedChangeCaptioningModel). While Phase 6 introduced hierarchical 2×2 tiling to improve spatial resolution, empirical analysis revealed two major domain-shift bottlenecks when applying the model to **Indian urban Google Earth imagery**:

1. **Unique Urban Morphology:** Indian cities exhibit dense, organic settlements and diverse rooftop materials (corrugated tin sheets, terracotta tiles, RCC slabs, asbestos) that differ dramatically from the planned suburban buildings in Western/Chinese datasets (LEVIR-CC). A frozen vision backbone frequently misinterprets reflective tin roofs as noise.
2. **Monsoonal Seasonal Invariance:** Indian satellite images undergo severe spectral color shifts between dry summer (brown soil) and lush monsoon (green foliage). Without targeted seasonal invariance, change captioning models hallucinate false structural construction/demolition whenever foliage changes color.

> ***CORE ENGINEERING OBJECTIVE:*** *Upgrade the project base to resolve patch arithmetic, eliminate seasonal false positives, correct class imbalance bias, and retain the full **Qwen2-VL-2B causal LLM decoder** via 4-bit NF4 quantization on ADA cluster compute nodes (with automatic GPU-to-CPU graceful fallback) while maintaining compatibility with src/dataset.py.*

## 2. The 6 Core Engineering Resolutions (The CodeAug Protocol)

To make the pipeline rigorous and mathematically sound, we resolve six critical concerns in the updated architecture:

| Engineering Concern | Previous Flaw / Ambiguity | Resolved Technical Specification |
|---|---|---|
| **1. Patch Grid Math & Pos-Embeds** | 256÷14 = 18.29 gives fractional edge patches; pos-embeds misaligned. | Enforce exact **252×252 px** input (252÷14 = 18, 18×18 = **324 clean tokens**). Bicubic-interpolate 2D pos-embed grid (16×16 → 18×18), excluding CLS. |
| **2. Decoder Backbone Routing** | Qwen2-VL-2B has a native vision tower; routing was ambiguous. | Explicitly **discard Qwen2-VL's native vision tower**. Instantiate as pure causal text LLM; inject RemoteCLIP change tokens via input projector. |

| | | |
|---|---|---|
| **3. Token Compression & VRAM** | MLP 1:1 projector feeds 324 tokens into LLM, causing KV-cache bloat. | Commit to **Q-Former (64 latent queries)**. Compresses 324 tokens $\rightarrow$ 64 tokens (**61.3% reduction**, sequence 424 $\rightarrow$ 164 tokens), fitting ADA memory. |
| **4. Imbalance Control** | Stacking 3:1 sampler + loss $\gamma$=2.0 double-counted change bias. | Use **single-lever 2:1 WeightedRandomSampler** with standard unweighted CE loss ($\gamma$=1.0). Prevents aggressive change-seeking against SFPR target. |
| **5. Bi-Temporal GLI Masking** | Independent masking missed brown-to-green transition pixels in $I_A$; called NDVI. | Use RGB **Green Leaf Index (GLI)**. Enforce **Bi-Temporal Union Mask** ($M_{veg} = M_A \cup M_B$), applying spectral jitter to transition pixels in both images. |
| **6. Evaluation Protocol** | Missing literature metrics; CIDEr noisy on small ~300 pair sets. | Report BLEU-1..4, METEOR, ROUGE-L, **LEVIR-CC Anchored CIDEr** (static IDF weights), Semantic Cosine Sim, SFPR (<5%), and SFNR (<8%). |

## 3. ADA Cluster Hardware Verification & GPU/CPU Execution Strategy

We verified your ADA cluster compute node profile (e.g., node gnode004) from historical SLURM logs in logs/phase.out. Importantly, we confirmed that **4-bit NF4 quantization (bitsandbytes) is officially supported on Compute Capability 6.0+ (Pascal CC 6.1)**. The only hardware limitation on Pascal is the lack of native bfloat16 (BF16) instructions.

| Hardware Layer | Verified Specification | Engineering Impact & Required Pipeline Rules |
|---|---|---|
| **GPU Node (gnode004)** | NVIDIA GeForce GTX 1080 Ti<br>**11 GB VRAM** (11,264 MiB)<br>Pascal Architecture (sm_61) | **CRITICAL FIX:** Retain **4-bit NF4 Quantization** for Qwen2-VL-2B and set **bnb_4bit_compute_dtype=torch.float16** (instead of bfloat16). Frozen ~1.54B causal text backbone (post-vision-tower pruning) consumes only ~1.2–1.5 GB VRAM, fitting comfortably in 11 GB. |
| **CPU Compute Node** | 16–32 x86_64 CPU Cores<br>36 GB+ System RAM allocated<br>No CUDA Acceleration | When CUDA is unavailable, run in **FP32 Full Precision** with multi-threading (OMP_NUM_THREADS=$SLURM_CPUS_PER_TASK) using system memory.<br>**Note:** SLURM scripts must explicitly request --cpus-per-task=16 (or 32) to override the 4-core default. |
| **Empirical VRAM Calibration** | Pre-Flight Script<br>check_vram_calibration.py | **Mandatory Verification Rule:** Before trusting any VRAM or timing assertion, run 1 real forward+backward step on gnode004 with torch.cuda.max_memory_allocated(). |
| **SLURM Constraints** | 4 CPU Cores / Job (Default)<br>Shared Multiprocessing Limits | All DataLoaders must enforce **num_workers = 0** to prevent shared memory deadlock on ADA nodes. |

### Why Retaining Qwen2-VL-2B via 4-Bit NF4 is Critical

By using bnb_4bit_compute_dtype=torch.float16, we avoid retreating to a from-scratch SimpleDecoder (Option A) or downsizing to a 0.5–1.5B model (Option B). Preserving the ~1.54B retained causal LLM backbone (verified via model.language_model.num_parameters() after pruning the native vision tower) is what enables the system to distinguish semantic 'seasonal vegetation shifts' from true 'structural building changes'. With 4-bit NF4 and FP16 compute dtype, the entire model fits comfortably within the 11 GB GTX 1080 Ti VRAM envelope.

## 4. Data Pipeline Compliance Analysis (src/dataset.py)

We reviewed the existing data pipeline in src/dataset.py. The table below demonstrates 100% backward compatibility and identifies the four modular additions required for Indian urban and seasonal change captioning:

| Pipeline Component | Existing Older Code (src/dataset.py) | Compliance Status & Required Modular Addition |
|---|---|---|
| **LEVIR-CC JSON Loader** | get_levircc_loaders() parsing LevirCCcaptions.json | **100% Compliant As-Is.** No changes to dictionary keys or format. |
| **Vocabulary & Tokens** | Vocabulary class (PAD=0, START=1, END=2, UNK=3) | **MODIFIED / SWAPPED OUT:** Because we commit to Qwen2-VL-2B as the causal LLM decoder, we replace the custom Vocabulary class with Qwen's native HuggingFace BPE Tokenizer (AutoTokenizer) and special-token scheme. |
| **Resolution & Crop** | Resizing to 224x224 or patch size 256 | **Modular Addition:** Add explicit **252×252 px** resizing in transforms_fn. |
| **Seasonal Vegetation** | Standard independent RGB color transforms | **Modular Addition:** Add BiTemporalUnionGLIJitter computing M_veg = M_A ∪ M_B via GLI > 0.05. |
| **Imbalance Sampling** | Standard RandomSampler / shuffle | **Modular Addition:** Add get_balanced_sampler() enforcing **2:1 changed-to-unchanged ratio**. |
| **CIDEr IDF Export** | On-the-fly TF-IDF corpus document frequency | **Modular Addition:** Export LEVIR-CC word IDF to checkpoints/levircc_cider_idf.pkl. |

## 5. End-to-End Two-Stage Training & Evaluation Roadmap

To prevent overfitting on the small Indian urban dataset (~200–500 pairs) while mastering change-captioning grammar, we execute a structured two-stage transfer learning pipeline:

| Stage & Dataset | Optimization Objectives & Parameters | ADA Cluster Execution & Verification |
|---|---|---|
| **Stage 1: Foundation Pre-training**<br>**LEVIR-CC (~10,000 pairs)**<br>**US/China Suburban Data** | • Train Q-Former + Visual LoRA (r=16) + Qwen2-VL-2B (4-bit NF4).<br>• 10 Epochs \| Effective Batch Size 16.<br>• LR: 1e-4 (Q-Former), 5e-5 (LoRA).<br>• Loss: Unweighted CE (γ=1.0) + InfoNCE (λ=0.1). | **GTX 1080 Ti (11 GB):** 4-bit NF4 with bnb_4bit_compute_dtype=torch.float16.<br>**Runtime Verification:** Measured empirically via pre-flight script. |
| **Stage 2: Indian Domain Adaptation**<br>**Indian Google Earth (~200–500 pairs)**<br>**Dense Cities & Monsoon Data** | • Pre-flight 15% val sweep for (τ_GLI, jitter_mag).<br>• Train LoRA & Q-Former for 15 Epochs \| LR: 2e-5.<br>• Enforce 2:1 WeightedRandomSampler.<br>• Apply Bi-Temporal Union GLI Jitter. | **GTX 1080 Ti (11 GB):** Fast few-shot adaptation.<br>**CPU Fallback Node:** Feasible backup if GPU queue is congested. |

## Balanced Evaluation & Diagnostic Protocol

On the held-out Indian test split (~50–100 pairs), the pipeline evaluates seven comprehensive metrics:
**1. Standard Literature Benchmarks:** BLEU-1, BLEU-2, BLEU-3, BLEU-4, METEOR, ROUGE-L, and **LEVIR-CC Anchored CIDEr**.
**2. Documented Tradeoff Note:** Anchored CIDEr provides high statistical stability on small test sets, but may inflate scores for Indian domain words (*tin, terracotta, RCC*) that are rare in LEVIR-CC.
**3. Domain & Seasonal Diagnostics:** Semantic Cosine Similarity (via SentenceTransformer), **SFPR** (Seasonal False-Positive Rate on unchanged seasonal pairs, target < 5%), and **SFNR** (Structural False-Negative Rate on changed pairs with color shifts, target < 8%).

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***Document Status:*** *IMPLEMENTATION READY* | ***Target Execution Script:*** *execute_phase6.sh* | ***Hardware Compatibility:*** *ADA Cluster GTX 1080 Ti (4-bit NF4 + FP16 compute dtype) & CPU Nodes*